

Assignment Submission Sheet

ARIN 310

Introduction to Artificial Intelligence

Unit 4 Assignment: Exploring Computer Vision and Image Recognition in Real-Life Applications

Matthew Kolonich

Professor Ulyana Martin

07/07/2025

Introduction Written Component:

I am going to look at 3 applications to help understand computer vision and recognition technologies that are used today in the AI and Machine Learning fields. I am going to look at Google Lens, which is a camera application that uses an algorithm model to determine objects within the picture. I will also explore This Person Does Not Exist(.com), which is an image generating application that creates images of people who do not exist. The last application I will look at is Remove Background Image. This is an application that removes background objects from pictures. All three of these applications use computer vision and recognition technologies in order to decipher images and pictures.

Computer Vision and image recognition technologies are one of the most interesting fields of AI and Machine Learning. “Computer Vision is a field of AI that enables computers and systems to derive meaningful information from digital images, videos, and other visual inputs — and take actions or make recommendations based on that information.”(DeepAI) This is literally a computer being able to visually see the world. It has been around since the 1950’s, but is really becoming an important AI technologies today. One of the biggest foundations of Computer Vision is the quickness at which it is processed. Part of Computer Vision is that the technology is currently able and working to better instantly process visual data as it is being seen. This instant processing is a huge factor in robotics, health, and many other fields because it allows data processing to be done in a much quicker manner.

The basic process of Computer Vision is that it takes a visual input and derives information from the visual input and then completes an action or recommendation based off this visual input. Computer Vision is usually using a combination of different types of learning. Generally, Computer Vision models are first taught supervised learning so they can be trained on visuals that are labeled. Then after the model has learned labeled models more unsupervised learning would be used. This would be loading massive data sets to the model in order for it to identify the variables that have already been obtained from the supervised learning. Computer Vision also sometimes use Reinforcement learning. This is where a model receives a reward or a demerit based on a decision. It will try to maximize its decision based off higher rewards. All three types of learning are used in Computer Vision to create models that can process data and make decisions instantly.

Computer Vision has some key terms to help understand some of its functions. These terms are object, model, segmentation, detection, tracking, counting, classification, and recognition. Object is the item being detected or recognized in images. Model is the trained algorithm that is identifying object classes. Segmentation which is the algorithm segmenting the object or image into different sections to extract meaningful data. The other terms more follow types of uses for Computer Vision. There is detection which is finding object in a frame. Tracking which follow objects or images through live video

feed. Counting which is counting objects or images within a frame. Classification which categorizes objects into categories. Recognition is identifying known objects or images. (Peterson)

Computer Vision is a field that is the way of the future. It can instantly look at an image and make a recommendation or conclusion from that image. This is helpful in almost every field or expertise whether it be health or business. Computer Vision has a way of analyzing data and images that humans never can. It can process large amounts of data faster and also has the ability, through special lens, to see and detect things that human eye never could. From these visuals decisions can be made in real time.

Google Lens Analysis Written Component:

The first application I looked at was Google Lens. Google Lens is an application that allows you to take a picture of an object and it will then try to identify the object. Google Lens gives many different recommendations and data about the object. One of the first things it tries to identify the object. It also compares it to similar objects and shows you items that can be bought that are the same object or similar. The AI also gives a rundown of the key features of the item. All in all the application has many great features because it allows you to identify, buy, learn, and compare any object you take a picture of.

I tried Google Lens on 5 objects and came away quite impressed. The first object I took a picture of was a lamp. Google Lens was able to identify it as a lamp as well as tell me the exact brand and type of lamp. It was 100 percent correct. I actually didn't even know the brand. The second object was a Peace Lily. Google Lens identified it as a Peace Lily immediately and gave some good growing tips. My third object was a Laz-a-Boy Chair. It identified the chair instantly and knew the type of Laz-a-Boy recliner it was. Google Lens did identify it as an electric recliner when it is not. This shows an issue in identification that Google Lens can make when determining specific details of an image. My 4th object was another chair. It was called the Pinecrest by Belfort. Google Lens was able to identify it as a chair, but was all over the place in what type. It definitely did not guess the right chair. It also had trouble identifying the color. It was artichoke and it thought it was gray or green. The last object was a glass blown vase that my uncle made. Google Lens identified it as a glass vase and then compared it to other glass vases that looked similar. All in all Google Lens I feel was highly accurate. It was able to identify each object and get some distinctive details about objects. Now while it did no some distinctive details, it was way off on some other objects specifics. I believe this is something that as more use and training is received, Google Lens will better be able to identify objects.

This application has a wide range of uses within many fields. It can be used for anything that needs identification. I think a good example could be in science or botany. I was able to identify my peace lily, but this would be good for out in the wild for identification as well. You could find plants that are safe to eat or are good for making a shelter. I think an industry this is huge for is retail. You can identify objects customers like and then instantly hand them other items that are similar or they are likely to buy. It is hard to give just 2 examples because Google Lens can be used for anything that requires identification.

Google Lens is a good example of Computer Vision. It is taken an object and analyzing the pixels within the object in order to create data variables about the image. It is then able to take these data variables and piece them together in order to come to its best identification of the object. Google Lens is also training itself with each new object it analyzes. It increases its database of knowledge with each picture. This is also reinforcement because the model can be trained on whether it was able to identify the object or not.

This would be the reward system that allows reinforcement learning to work. Google Lens uses Computer Vision to help its training and learning to improve its vision.

This Person Does Not Exist Analysis Written Component:

The next application I used was This Person Does Not Exist. This is an application that creates images of people who do not exist. I opened the webpage 5 times. The first image was of a woman in a hat with now background. The image looked very real and lifelike. The only think I can tell is the eyes look very dull and lifeless. The second image was a girl's face with a mountain in the background. The mountain on the left side was fine but on the right side of the girls head the mountain looked like a kaleidoscope. The third image was of a redheaded lady. The image looked normal except she had an earring in her left ear with a diamond and an earring in her right ear that was missing the diamond. My fourth image was of what seemed to be a 4 year old girl. Everything looked good including the background, but then I noticed her left ear was intermingled weirdly with her hair. The last image I got was of a man with glasses and grey hair. He looked normal except for his left ear was deformed.

The application does a good job of getting a picture of a person who does not exist. It does a really good job of creating the face portion of the picture. The further the image gets away from the center of the face the more the image seems to have a chance of being distorted. "Sometimes, the datasets may not combine with each other perfectly leaving some visible flaws in the image."(Singh) I believe this could be because of the sheer amount of data points. Hair seems to mess up a lot with ears and its look in general. This could be because hair can go in billions of different ways and it is hard to analyze and compute so many different combinations. This helps to create images that might be off in the background and around the edges. The realism seems to be really accurate for the face. This probably means it has more data points available to learn and train with of the face.

There are many applications that could be used for this. I see one big one is retail. Creating images of people that could be used in ads, but then not having to use resources to pay models. This could be the same in other fields as well, such as entertainment. I also tend to take a negative look on this and "People can use these fake images and hide their identity online for malicious use" (Singh). This could be creating a picture of a child to try to collect money from a long lost relative that doesn't exist. Mainly I see this as an application that could be used in fraud. I see it as an application that can create images of people for learning purposes. This is a good application for showing the explainability and transparency of Computer Vision. I do not see a positive application for the actual images except to use to understand process behind the technology.

Remove.bg Application Analysis Written Component:

Remove Background Image is an application that removes the background of images. You can then add new backgrounds and designs in the background of the object. I attempted 3 images. I first used a picture of myself sitting at the computer. This image was pretty straightforward and had no issues with removing the background. I next tried a couple of pictures with multiple subjects. I was really impressed. I thought it would have trouble and cut out 1 of the subjects. Instead it included all the subjects. My example was 3 guys sitting on a big couch far apart. It ended up keeping the couch and all 3 subjects and removing everything else. The last image I used was a complex image. I used a picture of a skyscraper surrounded by other skyscrapers of the same color. The application could not remove a background and requested I used a magic brush or send the image in for investigation. All in all the application did a great job of being able to identify what part of the image needed to be removed.

The accuracy of this application was pretty amazing. I was really quite surprised that it was accurately able to remove the background in multiple subject pictures immediately. The benefits has a wide range of possibilities. Advertisement is a huge area. It can help to create images for commercial interests and ads. "Ecommerce photography can pose some challenges: even in the most controlled studio environment, blemishes are inevitable; supposedly white backgrounds are never totally white."(Canva) This shows how it can be used to create pictures for advertisements that do not have to worry about background interference. All of the attention is only on the product. It can also be used for privacy and security matters. Today much communication is done through video and you might not want someone to see where you are or what is around you. Remove Background Image can do this for you. Another benefit for marketing and advertising is maximizing backgrounds in your images. For example if you were selling a car you might wish to put the car in front of a dealership or sitting on a cliff, anything to entice people about the image. This can also be a great in design fields. You can try to determine how different things would look. An example is what type yard design around your house. This application could easily come up with many alternative options for your yard design. I believe this application has great uses in design and advertisement, as well as many other uses.

I do see some limitations with the application. The main one being with complex images. These images usually do not have a clear defined subject as well as being similarly colored or designed throughout the image. This makes it difficult for the algorithm to determine what should be removed in the background. When there are complex images the algorithm cannot define, you can send them the image to be contributed to help train. "When you upload an image we do not use this image to train the AI."(Canva) This also shows that this algorithm is mainly using supervised learning. It is only training from pictures that have been contributed. These contributed photos are then classified and then used to train the algorithm making this supervised learning.

This application really impresses me. It is amazing that the Computer Vision used in this algorithm can determine the background so easily. It really makes me want to learn more about the variables that the algorithm is using to determine background. I think with so many different types of images there are probably many different variables in order to determine what is and what is not in the background.

Ethical Considerations Analysis Written Component:

Computer Vision is a technology that is being used with greater frequency within our world. It is being used in almost all fields and areas of life. That being said brings up ethical implications about using this technology in the real world. “Who controls the ownership of images and videos depicting you or your property? How should they be used—if at all? And how do individuals provide consent for the use of personal data or even discover where it is being stored?” () This is one of the biggest concerns with the use of Computer Vision technology. A picture can be taken of anything, but who truly owns the rights to that picture. An example would be someone taking a picture of a lady, then using it in an ad without her consent or knowing. Should the owner of the picture or the person the picture is of have rights to the image? Besides privacy and personal property another big issue can be fraud. Images can be used to deceive people in to believing things that are not true. This has been one of the biggest ethical concerns concerning Computer Vision. Computer Vision can also effect society by changing how we do things in general. An example is the self-driving car. What are the implications of humans no longer driving? What is done with that time? Is the car driving accurately? These could all have huge societal effects.

The ethical considerations I can see from Google Lens are mainly through who is using the application. Google Lens in general is identifying objects, but what if the person is using it for unethical purposes. A thief could use it to identify something he wishes to steal or the value of something. I also see ethical implications on society. Google is trusted and when people use the application I believe in general will trust it, but just like my chair it might get the information wrong. This can be an ethical implication on society by causing people to think certain things that in fact might not be true.

This Person Does Not Exist has much more ethical considerations for me. I see this as a tool that can be used to deceive and trick people. It can create an image of someone who doesn't exist, which then could be used to convince someone the person is real to use for some malicious intent. “It's fair to be worried about any new technology. This website creates faces that no one has ever seen before, making it a very useful tool for malevolent users. People can use these fake images and hide their identity online for malicious use.”(Singh) As you see even the creators at the application see the potential for abuse. I also think about societal implications. How would it affect people knowing they are watching people who do not exist? My example, would watching actors who are not real have an effect on people who are watching the shows?

Remove Background Image has ethical concerns as well. It too has the potential to create fraud or monovalent intent. This could be changing where something is located. You could change the background of a house so it looks like it's at a river instead of a field. Another example would be showing someone of you pictures in Spain even though you never left your room. I could also see this

being used to show where someone isn't at. An example would be someone kidnapped or runaway. They could show they were at the Grand Canyon when they are in Florida. This are some wild extreme examples, but show the ethical concerns one can have with Remove Background Image.

Conclusion Written Component:

Google Lens, This Person Does Not Exist, and Remove.bg are all applications that are using Computer Vision. All three applications show how Computer Vision is used in the real world today and can help to decipher images and objects. They are all using the objects they see to create variables of data in order to decipher and adapt other future objects they see. These all have a big impact on the world today.

Google Lens is highly effective in many different sectors. Construction is one that it can be used for. Construction sites have many coming and going people. Google Lens can help to identify people on the construction site. It could also be used for safety by identifying whether all workers have on safety equipment. Another sector is in manufacturing. It can be used to identify needed material, right type of parts, or defects on the product. These are just a few of the sectors or applications for Google Lens.

This Person Does Not Exist is highly effective for advertisement. You can create a model without having to pay them or without offending anybody. This is also good for marketing because you can create an agent that appeals to everyone. They know people respond better to certain looks and styles of other people, so this could be used to find the best images of people to sell products.

Remove.bg can be used for many different sectors. The two I see it the most is content creation and advertising. In content creation you are trying to entice as many people as possible into looking at your images. By removing the background and creating a more interesting one you are creating more viewership of your content. In advertising you can use the background image to help sell product. Having a volleyball set shown in a backyard will sell, but if you change the background to a beach will it sell better? With advertising you can use Remove.bg to put your product anywhere in the world.

I see the future as being bright with Computer Vision. I believe it will lead to more robotic features in life. From the Roomba to the self-driving cars Computer Vision is changing our society and how we do things. Many different facets in life can be done in a less time consuming or energy consuming manner. An example is the self-driving cars. You are able to do something while the car is driving itself. I also see the future being impacted with better 3-D visualization which should help with all programs that need to understand spatial awareness. This is especially true in robotics, such as the arm that needs to pick up an object. Another future advantage of Computer Vision is Hyperspectral imaging. This is using lens that can pick up different types of visual information that the human eye cannot. This could be radiation levels, UV rays, or any other type of imaging that the human eye cannot see. This hyperspectral imaging could be useful in healthcare by identifying diseases, illnesses, or sicknesses quickly and accurately. This could also be used to identify light rays in the color spectrum from space.

The human eye cannot see these light rays, but Computer Vision could. Computer Vision is the ability for computers to analyze their visuals and images and come to conclusions and recommendations based off these images. Computer Vision is one of the biggest technologies in AI and the future.

References

Canva Austria GmbH. (n.d.). *Remove.bg – Help Center: General & Website*.
<https://www.remove.bg/help/general-website>

DeepAI. (n.d.). *Computer vision*. <https://deepai.org/machine-learning-glossary-and-terms/computer-vision>

Kaur, J. (2024, October 22). *Ethical Considerations and Bias in Computer Vision (CV)*. XenonStack.
<https://www.xenonstack.com/blog/ethical-considerations-in-computer-vision>

Singh, D. N. (2022, May 25). *ThisPersonDoesNotExist(.com). The person in the image does not exist*. Medium. <https://medium.com/@D.N.S/thispersondoesnotexist-com-42767d242d0>

Intel Corporation. (n.d.). *What is machine vision?* Intel.
<https://www.intel.com/content/www/us/en/learn/what-is-machine-vision.html>

Peterson, N. (2025, July 7). *History of computer vision principles*. alwaysAI.
<https://alwaysai.co/blog/history-computer-vision-principles>